Program / Source Code

```
#include <stdio.h>

#define FIXED_CHARGE 3000
#define RATE1 225
#define RATE2 450
#define MAX_READING 65535

int main()
{
    int previous, current, units;
    long long bill;

    printf("Enter previous reading: ");

    if (scanf("%d", &previous) != 1)
    {
        printf("Invalid input");
        return 0;
    }

    printf("Enter current reading: ");

    if (scanf("%d", &current) != 1)
    {
        printf("Invalid input");
        return 0;
    }

    if (previous < 0 || previous > MAX_READING ||
        current < 0 || current > MAX_READING)
    {
        printf("Invalid meter reading");
        return 0;
    }

    if (current >= previous)
        units = current - previous;
```

```
    else if (previous == 65535 && current == 0)
        units = 1;
    else
    {
        printf("Invalid meter rollover");
        return 0;
    }

    if (units <= 100)
        bill = FIXED_CHARGE;
    else if (units <= 200)
        bill = FIXED_CHARGE + (units - 100) * RATE1;
    else
        bill = FIXED_CHARGE + 100 * RATE1
             + (units - 200) * RATE2;

    printf("Units Consumed = %d\n", units);
    printf("Total Bill = Rs.%lld.%02lld\n",
           bill / 100, bill % 100);

    return 0;
}
```

Output Screenshot



Output Screenshot



Program / Source Code

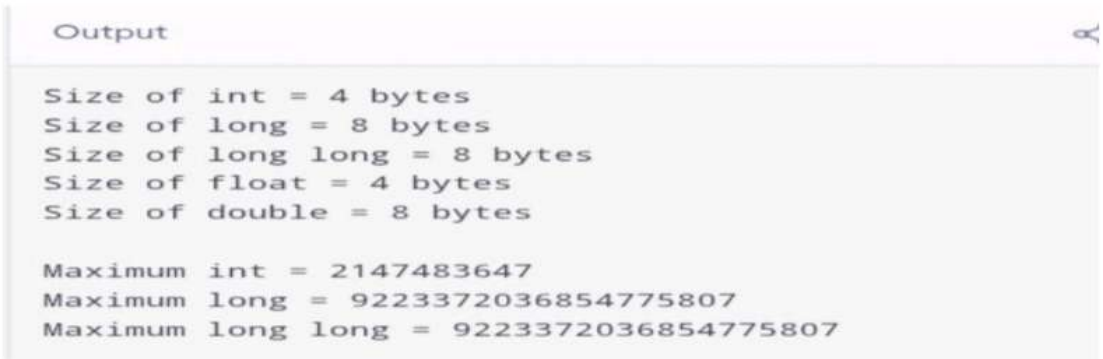
```
#include <stdio.h>
#include <limits.h>

int main()
{
    printf("Size of int = %zu bytes\n", sizeof(int));
    printf("Size of long = %zu bytes\n", sizeof(long));
    printf("Size of long long = %zu bytes\n",
           sizeof(long long));
    printf("Size of float = %zu bytes\n", sizeof(float));
    printf("Size of double = %zu bytes\n", sizeof(double));

    printf("\nMaximum int = %d\n", INT_MAX);
    printf("Maximum long = %ld\n", LONG_MAX);
    printf("Maximum long long = %lld\n", LLONG_MAX);

    return 0;
}
```

Output Screenshot



Program / Source Code

```
#include <stdio.h>

#define FIXED 3000
#define RATE1 225
#define RATE2 450

int main()
{
    int units;
    long long bill;

    printf("Enter consumed units: ");
    scanf("%d", &units);

    if (units < 0)
    {
        printf("Invalid units");
        return 0;
    }

    if (units <= 100)
        bill = FIXED;
    else if (units <= 200)
        bill = FIXED + (units - 100) * RATE1;
    else
        bill = FIXED + 100 * RATE1
             + (units - 200) * RATE2;

    printf("Total Bill = Rs.%lld.%02lld",
           bill / 100, bill % 100);

    return 0;
}
```

Output Screenshot



Program / Source Code

```
#include <stdio.h>

#define PEAK_START 18
#define PEAK_END 22

int main()
{
   long long normalWh, peakWh;
   long long normalRate, peakRate;
   long long bill;

   printf("Enter normal energy in Wh: ");
   scanf("%lld", &normalWh);

   printf("Enter peak energy in Wh: ");
   scanf("%lld", &peakWh);

   printf("Enter normal rate in paise/kWh: ");
   scanf("%lld", &normalRate);

   if (normalWh < 0 || peakWh < 0 ||
      normalRate < 0)
   {
      printf("Invalid input");
      return 0;
   }

   peakRate = normalRate * 120 / 100;

   bill = (normalWh * normalRate +
         peakWh * peakRate) / 1000;

   printf("Total Bill = Rs.%lld.%02lld\n",
         bill / 100, bill % 100);

   return 0;
}
```

Output Screenshot;

